Where do the other worlds go? A toy-model test of Born-rule filtering in a local Ising chain

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Abstract

Strasberg and collaborators reported, in random matrix toy models, a filtering effect in which Born-atypical frequency classes of quantum histories recohere while Born-typical ones remain decoherent. We test whether the same qualitative pattern appears in a physically motivated system: a 9-qubit transverse-field Ising chain ($D = 512$) with local, nearest-neighbor couplings and exact unitary evolution. In our model, the recoherence parameter ε dips near the Born frequency ($n_1/L \approx p_1 = 0.746$) with a $3.1\times$ gap over the combinatorial peak, averaged over 10 Haar-random initial states, matching the qualitative shape of the random matrix prediction. At the individual-branch level ($L = 12$, 4096 branches), only 451 branches remain approximately decoherent ($\varepsilon < 0.3$), and these cluster at Born-typical frequencies. In this toy model, the Born-typical filtering pattern emerges from the interplay of finite dimensions and local unitary dynamics—though whether this generalizes beyond toy models remains open.

1 Introduction

Quantum mechanics has a puzzle at its core, and it is surprisingly easy to state.

The Schrödinger equation is deterministic and linear. If you know the state of a system right now, you can compute its state at any future time—no randomness, no choices, no collapse. And yet, every time we open the box and look at the cat, we see one definite outcome. The math says the cat is in a superposition of alive and dead. Our eyes say it is one or the other.

This is the measurement problem, and it has been with us since the 1920s. There is no shortage of proposed solutions. The Copenhagen interpretation says the wave function collapses upon measurement, but never quite explains what counts as a measurement. Bohmian mechanics adds hidden variables. Objective collapse theories add new physics. And then there is Everett's proposal [Everett, 1957]: nothing collapses at all. The Schrödinger equation is the whole story. What we experience as "collapse" is the universe splitting into branches—every possible outcome happens, each in its own branch of reality, and we simply find ourselves in one of them.

The Everett picture is elegant, but it raises its own question. If every outcome happens, why do we observe outcomes with the frequencies predicted by the Born rule? If there is one branch where the electron goes left and one where it goes right, and both are equally real, why do we not see 50/50 statistics regardless of the quantum state?

This is sometimes called the *probability problem* of the many-worlds interpretation, and it is the subject of a large literature. Deutsch [Deutsch, 1999] gave the first decision-theoretic argument: a rational agent in a branching universe *should* weight branches by their squared amplitudes. Wallace [Wallace, 2012] refined and extended the proof; Carroll [Carroll, 2019] offers an accessible account and argues, with Sebens, that Born's rule follows from self-locating uncertainty at the moment of branching. Zurek [Zurek, 2005] pursued a different strategy, deriving Born's rule from environment-assisted invariance ("envariance") without decision theory. These approaches are persuasive to some and unconvincing to others. What they share is that they introduce additional postulates—rationality axioms or symmetry principles—beyond the Schrödinger equation itself.

Recently, Strasberg and collaborators [Strasberg et al., 2024, Strasberg and Schindler, 2023, Strasberg et al., 2026] have explored a different route. Their argument is physical, not decision-theoretic. In a finite-dimensional Hilbert space, the histories of a quantum system cannot all remain decoherent as the number of branching events grows. There are more histories than dimensions, so branches must start overlapping. In random matrix toy models, they reported that this overlap is not uniform: the histories whose frequencies match the Born rule remain decoherent, while the others do not. Whether this filtering pattern is a generic feature of finite-dimensional quantum mechanics or specific to the models studied remains an open question.

Strasberg's results are established for random matrix Hamiltonians and related asymptotic arguments. We ask: does the same qualitative behavior appear in a concrete, physically motivated Hamiltonian? We take a 9-qubit transverse-field Ising chain—a nearest-neighbor spin model near its critical point—and check.

2 The Puzzle: All Branches Exist, but the Statistics Are Wrong

Take a loaded coin: 75% heads, 25% tails. We know what probability means here. Flip it many times, and about 75% of the flips will come up heads. That is what we observe. That is what we mean by the coin being "loaded."

Now flip it once in the Everett picture. Both outcomes happen. There are two branches: one where you got heads, one where you got tails. Both are real. Both exist. So what does "75% probability" even mean, if both outcomes occur with certainty?

Flip again. Now there are four branches: HH, HT, TH, TT. All four exist. Of these, one has 0 heads, two have 1 head, and one has 2 heads. If you just count branches, the most common outcome is 50% heads—not 75%.

After three flips, there are 8 branches. The branch-counting peak is still at 50% heads. After ten flips, there are 1024 branches, and $\binom{10}{5} = 252$ of them have exactly 50% heads, while only $\binom{10}{7} = 120$ have 70% heads. The more you flip, the worse it gets: branch counting says 50%, but we observe 75%.

This is the probability problem of the many-worlds interpretation. If every branch is equally real, what does probability mean? And why do we see the loaded-coin statistics instead of the branch-counting statistics?

3 Background: Recoherence in Finite Hilbert Space

The key is that Hilbert space is finite.

A quantum system with m qubits lives in a Hilbert space of dimension $D = 2^m$. Think of this as a pond with a fixed surface area. Each branch of the wave function is a lily pad. After L coin flips, there are 2^L lily pads. When L is small, there is plenty of room—each pad has its own patch of water, and they do not interfere. But the pond is finite. When $2^L > D$, the pads are competing for the same surface. They crowd, overlap, and shade each other out. Not all can survive as distinct plants.

In quantum language, branches that "overlap" are no longer orthogonal. Their quantum states point in nearly the same direction, which means they cannot be distinguished by any measurement. They have lost their identity as separate worlds. This language draws on the decoherent (or consistent) histories framework [Griffiths, 1984, Gell-Mann and Hartle, 1990], in which a set of histories counts as "classical" only when their mutual overlaps are negligible. Following Strasberg [Strasberg et al., 2026], we call the loss of this condition *recoherence*—the opposite of decoherence.

Strasberg and collaborators [Strasberg and Schindler, 2023, Strasberg et al., 2026] reported that, in their random matrix models, this competition is not democratic. The branches that get shaded out are not random. They are the ones near 50% heads—the branch-counting peak. The branches near 75% heads—the Born frequency—remain more nearly decoherent.

3.1 Measuring Decoherence

To make this precise, we need a way to measure how much two branches overlap.

We group branches by their frequency—the total number of heads n_1 out of L flips. The frequency-class state $|\psi(n_1)\rangle$ is the quantum state obtained by adding up all branches that got exactly n_1 heads:

$$|\psi(n_1)\rangle = \sum_{\mathbf{x}:\, n_1(\mathbf{x})=n_1} \Pi_{x_L} U_L \cdots \Pi_{x_1} U_1 |\Psi_0\rangle, \tag{1}$$

where U_k is the evolution between flips and Π_{x_k} picks out the heads-or-tails outcome at flip k.

The *recoherence parameter* ε [Strasberg and Schindler, 2023] measures how much a frequency class overlaps with any other frequency class—a score from 0 to 1:

$$\varepsilon(n_1) = \max_{n_1' \neq n_1} \frac{|\langle\psi(n_1')|\psi(n_1)\rangle|}{\sqrt{\langle\psi(n_1)|\psi(n_1)\rangle \cdot \langle\psi(n_1')|\psi(n_1')\rangle}}. \tag{2}$$

At $\varepsilon = 0$, the frequency class is fully decoherent (orthogonal to all others). At $\varepsilon = 1$, it is fully recoherent (indistinguishable from another class).

3.2 The Prediction

In their random matrix toy models, Strasberg and collaborators found that, as the number of flips L grows:

- Frequency classes near 75% heads have low ε—they stay decoherent,

- frequency classes near 50% heads have high ε—they blur together,
- the gap grows with L.

The pond is too small for all the lily pads near 50%. They shade each other out. The ones near 75% remain more nearly decoherent.

4 Numerical Experiment

We simulate repeated coarse-grained measurements on a 9-qubit Ising chain and analyze the decoherence structure of the resulting branches.[1]

The setup can be thought of as a loaded quantum coin flip repeated L times. The coin comes up heads 75% of the time. In the many-worlds picture, every outcome sequence occurs as a branch. We track all 2^L branches and compute the recoherence parameter ε for each, asking: which branches remain approximately decoherent, and which lose their identity through overlap with neighbors?

4.1 Setup: The Ising Chain

We use 9 qubits arranged in a line, interacting through the transverse-field Ising Hamiltonian:

$$H = \frac{J}{2}\sum_{i=0}^{7} Z_i Z_{i+1} + h_x \sum_{i=0}^{8} X_i, \tag{3}$$

with parameters $J/2 = 0.5$ and $h_x = 0.9045$, placing the system near the quantum critical point ($h_x/J \approx 0.9$). In this regime, the nearest-neighbor couplings generate significant entanglement and spreading of correlations. The Hilbert space has dimension $D = 2^9 = 512$.

Each "coin flip" works like this: let the qubits evolve under H for one time step, then ask a coarse question about the result. Our question is: are at least 4 of the 9 qubits in state $|1\rangle$? If yes, we call it "heads." If no, "tails." Since 382 of the 512 basis states have Hamming weight ≥ 4, the coin is loaded: $p_{\text{heads}} = d_1/D = 382/512 \approx 75\%$, where d_1 is the rank of the "heads" projector. In Strasberg's framework, this ratio d_1/D defines the Born-typical frequency: it is the single-trial Born probability averaged over Haar-random initial states. We verified numerically that our Ising chain produces effective single-step probabilities consistent with this ratio ($p_1^{\text{eff}} = 0.747 \pm 0.018$ over 200 Haar-random states).

No actual measurement is performed. The system evolves unitarily, and we apply the heads/tails decomposition analytically—projecting the state vector onto the two subspaces without collapsing it. The 9-qubit system is both the coin and the recorder. There is no separate environment.

4.2 Methods

Each evolution step applies $U = e^{-iH\Delta t}$ with $\Delta t = 1.2$, computed via exact matrix exponentiation.

Initial states are drawn Haar-randomly on the full D-dimensional Hilbert space. All frequency-class results are averaged over 10 independent random seeds; error bars show $\pm 1\sigma$.

[1] All simulation code is available at https://github.com/eschnou/quantum-recoherence.

Branch-level decoherence counts use the operational threshold $\varepsilon < 0.3$. This threshold is a conventional choice; the qualitative conclusion—that survivors cluster at Born-typical frequencies—holds for thresholds in the range $[0.1, 0.5]$, though the absolute count of "decoherent" branches varies.[2]

4.3 Frequency-Class Results

We first analyze frequency classes (groups of branches with the same number of heads), which is cheap enough to run at $L = 15$ flips. We repeat with 10 different random starting states. For each run, we compute ε for every frequency class.

Figure 1 shows the result. The frequency classes near 75% heads (the Born prediction) have low ε—they remain approximately decoherent. The classes near 50% heads (the branch-counting peak) have ε approximately three times higher—they overlap significantly with neighboring classes.

As a sanity check, we also run Strasberg's own random matrix model at the same dimension and coin loading. The two curves match: the Ising chain reproduces what random matrix theory predicts.

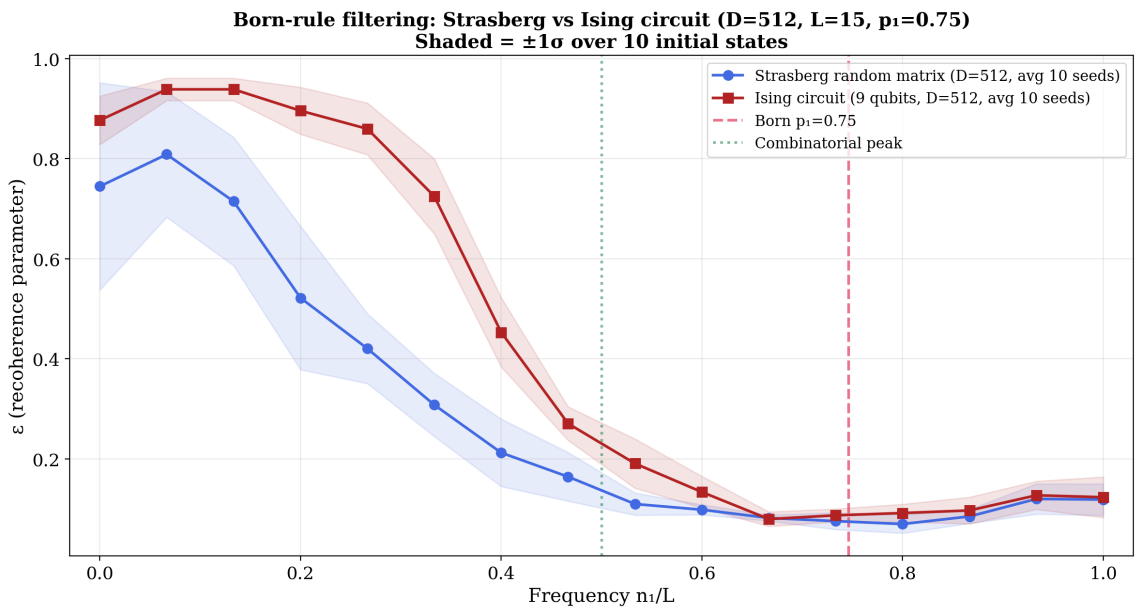

Figure 1: Recoherence parameter $\varepsilon(n_1/L)$ for the 9-qubit transverse-field Ising chain (red, exact matrix exponentiation) and Strasberg's random matrix model (blue) at $D = 512$, $L = 15$, $p_1 = 0.746$. Averaged over 10 Haar-random initial states; shaded regions show $\pm 1\sigma$. Both curves dip near the Born frequency (red dashed) and rise toward the combinatorial center (green dotted), showing the same qualitative filtering pattern.

The result is robust. All 10 seeds show the same qualitative shape. The standard deviation of ε across seeds is small relative to the Born–combinatorial gap, indicating that the effect is a property of the Hamiltonian, not of special initial states.

[2] Robustness checks (threshold sensitivity, Δt variation) are included in the code repository.

4.4 Branch-Level Analysis

So far we have grouped branches by their frequency (how many heads total). But we can go deeper: track every single branch individually. Each specific sequence of heads and tails—like HHTHTTHHTTHH—is one branch, one possible history of the coin. We compute ε for each one: how much does this branch overlap with any other?

Figure 2 shows what happens as we increase the number of flips. At $L = 4$ (16 branches, 512 dimensions), the pond is nearly empty—almost every branch is a distinct world. At $L = 9$ (512 branches, 512 dimensions), the pond is full. At $L = 12$ (4096 branches, 512 dimensions), it is eight times oversubscribed. Only 451 branches remain approximately decoherent ($\varepsilon < 0.3$). By $L = 13$ (8192 branches), 565 survive—slightly above D, since the threshold $\varepsilon < 0.3$ allows partial overlap.

The survivors cluster at Born-typical frequencies.

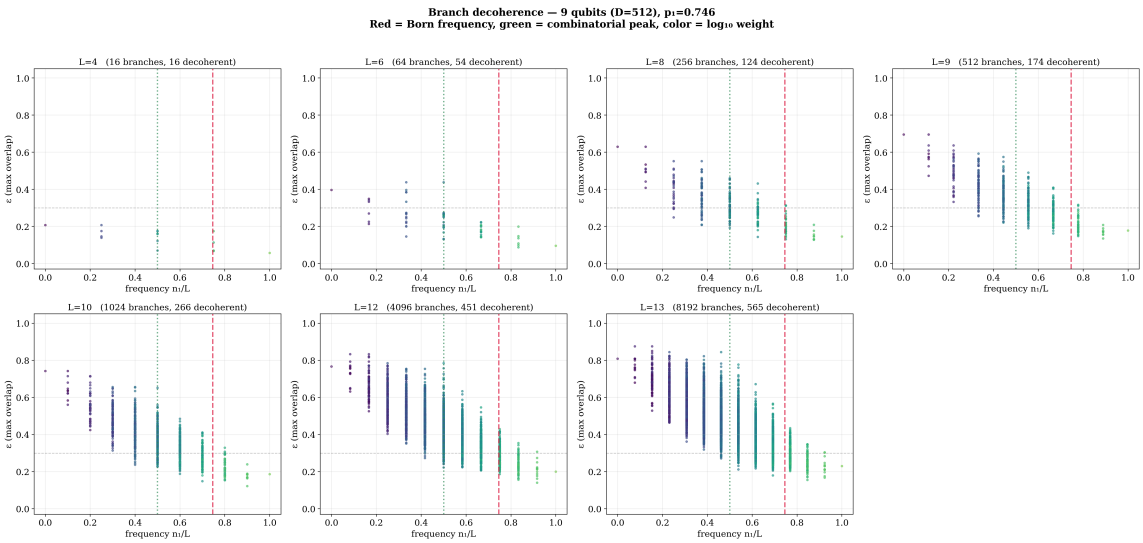

Figure 2: Recoherence parameter for individual branches at seven values of L. Each dot is one of 2^L branches; color encodes log weight. The decoherence threshold $\varepsilon = 0.3$ is shown as a gray dashed line. At $L = 13$, 8192 branches compete for 512 dimensions; only 565 satisfy $\varepsilon < 0.3$, clustered near the Born frequency (red dashed).

Figure 3 makes the census explicit. The total number of branches grows exponentially (2^L), but the growth of decoherent branches slows dramatically around $D = 512$. The dimension D is not a hard ceiling—approximate decoherence ($\varepsilon < 0.3$) allows more than D coexisting branches—but it sets the scale at which the competition intensifies.

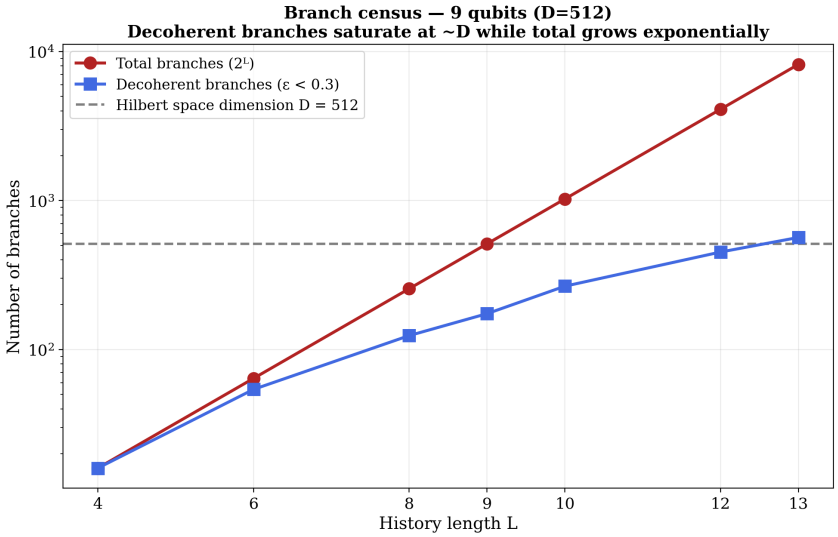

Figure 3: Branch census: total branches (red) versus decoherent branches with $\varepsilon < 0.3$ (blue). Growth slows dramatically around $D = 512$ (black dashed), while total branches grow as 2^L.

Figure 4 shows the frequency distribution of the survivors. The distribution of decoherent branches matches a binomial with Born probability $p_1 = 0.746$, not the combinatorial distribution $\binom{L}{n_1}/2^L$ that would arise from naive branch counting. The process is not literally i.i.d.; the binomial arises as the Born-rule prediction for the frequency distribution of surviving branches.

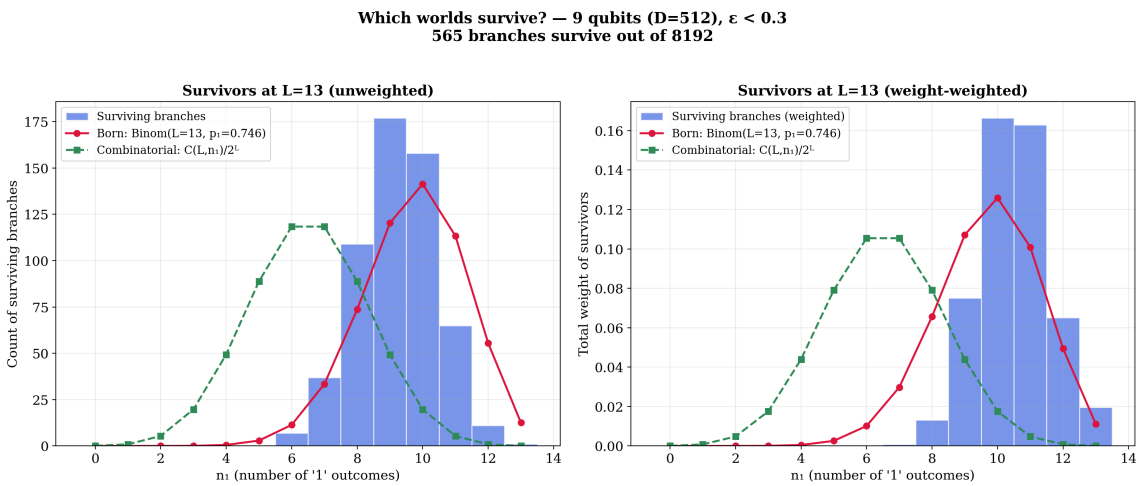

Figure 4: Frequency histogram of surviving branches at $L = 12$. Left: unweighted count. Right: weighted by branch norm. The red curve shows the Born binomial $\mathrm{Binom}(L, p_1)$; the green curve shows the combinatorial distribution. Survivors follow Born, not branch counting.

4.5 The All-Zeros and All-Ones Asymmetry

Two extreme branches tell the story most vividly.

The *all-tails* branch—tails on every single flip. This is the most anti-Born history possible. Its weight is 2×10^{-8} (essentially zero), and its $\varepsilon = 0.77$—it overlaps heavily with

neighboring branches and fails the approximate decoherence criterion by a wide margin.

The *all-heads* branch—heads every time. There is only one such history out of 4096, so by branch counting it is just as rare as all-tails. But the coin is loaded 75% heads, so this outcome is not absurd. Its weight is 0.025 (a million times larger than all-tails), and its $\varepsilon = 0.20$—it remains approximately decoherent. It has no siblings with the same frequency to compete with, so it retains its share of the Hilbert space.

In this model, anti-Born branches do not just have low weight. They also fail the approximate decoherence criterion used to identify distinct record-bearing histories.

4.6 The Gram Matrix

Figure 5 shows the overlap between every pair of branches at $L = 10$ (1024 branches), arranged by frequency. Dark means low overlap; bright means high overlap. The Born-frequency blocks are dark—those branches remain approximately orthogonal. The combinatorial-peak blocks are bright—those branches overlap significantly.

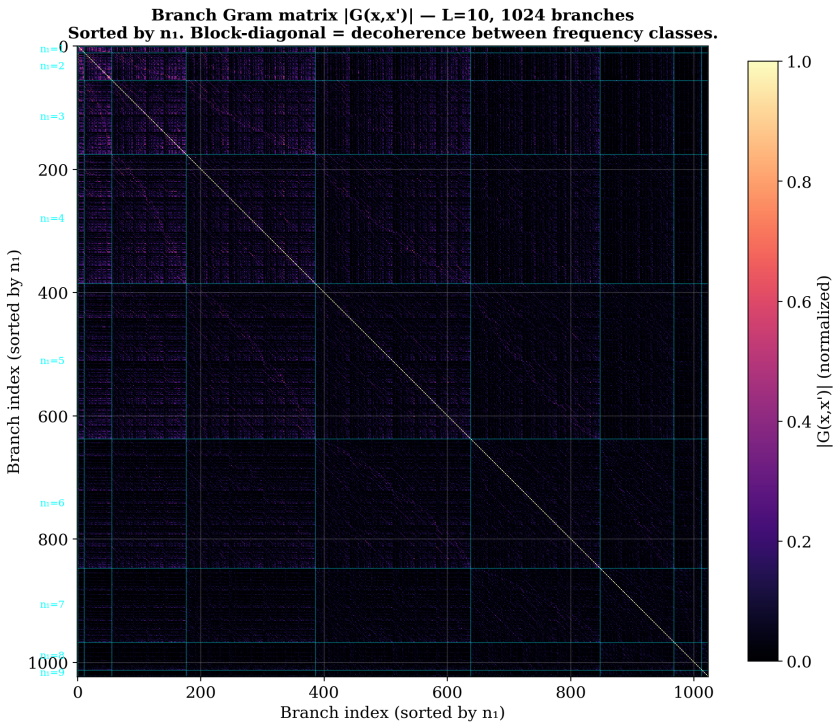

Figure 5: Absolute value of the normalized Gram matrix for 1024 branches at $L = 10$, sorted by n_1. Block-diagonal structure reflects frequency-class grouping. Brighter off-diagonal elements indicate recoherence. The central (combinatorial-peak) blocks are more coherent than the Born-frequency blocks.

4.7 Individual Branches

Finally, we pick four individual branches and trace their evolution step by step (Figure 6).

Each row shows one branch's outcome sequence, how its running frequency $n_1(k)/k$ evolves flip by flip, and its weight $\|\psi(\mathbf{x}, k)\|^2$ at each step. The Born-typical branch sees its frequency converge to 75% and retains significant weight. The anti-Born branches drift toward 50% and their weight drops by many orders of magnitude.

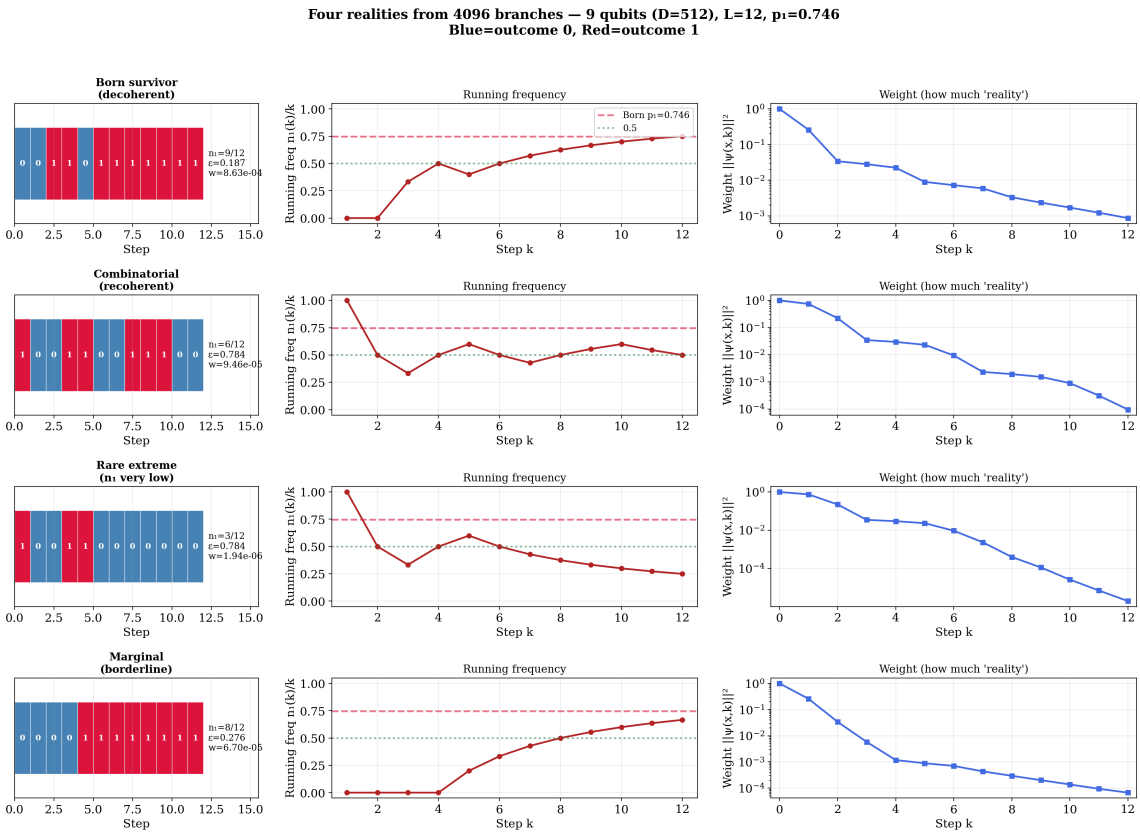

Figure 6: Four representative branches at $L = 12$. Each row shows: the outcome sequence (blue = 0, red = 1), the running frequency $n_1(k)/k$ converging (or not) to p_1, and the branch weight $\|\psi(\mathbf{x}, k)\|^2$ at each step.

5 Limitations

This work has clear limitations.

- **Small system.** Nine qubits ($D = 512$) is tiny. Real physical systems have Hilbert spaces of exponentially large dimension. We cannot say whether the $3.1\times$ gap grows, shrinks, or stays constant as D increases. Strasberg's analytical results [Strasberg et al., 2024] suggest it should improve, scaling as $D^{-\alpha}$, but we have not verified this numerically.

- **Classical simulation only.** We simulate the full state vector on a classical computer. This limits us to small systems. A quantum computer could in principle probe larger Hilbert spaces, but the analysis requires tracking exponentially many branches, which is inherently hard.

- **Not a proof.** We demonstrate the effect for one Hamiltonian (transverse-field Ising) with one coarse-graining (Hamming weight). We have not proven that Born-rule filtering occurs for all Hamiltonians or all reasonable coarse-grainings.

- **Specific coarse-graining.** The Hamming-weight threshold ≥ 4 gives $p_1 = 0.746$. Different thresholds give different p_1, and we have not systematically explored the dependence on the coarse-graining.

- **Time-step dependence.** The Born–combinatorial gap depends on Δt, requiring sufficient evolution between projections for the dynamics to spread correlations. The reported $3.1\times$ gap is specific to $\Delta t = 1.2$.
- **Integrable dynamics.** Our transverse-field Ising Hamiltonian is integrable (solvable by Jordan-Wigner transformation), unlike the chaotic random matrix models used by Strasberg. The filtering pattern appears nonetheless, suggesting the mechanism may not require chaos—but this also means we cannot claim to have tested it in a chaotic regime.
- **No environment.** Real decoherence involves tracing out an environment. Our system has no environment—the 9 qubits are the whole universe. This is intentional (it matches Strasberg's formalism, where the system itself is the recorder), but it is a simplification.

6 Interpretive Discussion

The numerical results in Section 4 stand on their own. This section offers speculative interpretive commentary—our reading of what the filtering mechanism might mean if it generalizes beyond toy models. These are philosophical proposals, not consequences of the simulation alone, and they go beyond what Strasberg and collaborators claim in the attached references.

6.1 Collapse as Identity, Not Process

In the standard story, wave function collapse is something that *happens*: at the moment of measurement, the superposition is somehow reduced to a single outcome. The Everett picture removes this event but replaces it with a question: why does each branch-observer see Born-rule statistics?

Our toy model suggests a possible reframing. In the simulation, the coarse-graining (the "observer") is a fixed projection—a particular way of carving up the Hilbert space into "outcomes." The system evolves unitarily. The projections define what counts as a branch. And the finite geometry of the Hilbert space constrains which branches can coexist as approximately decoherent histories.

If this picture generalizes, collapse would not be a process that happens at a moment in time, but a statement about which patterns of outcomes can be sustained as separate, non-interfering histories in a finite-dimensional space. In our 9-qubit model, the Born-typical patterns survive and the anti-Born patterns do not.

6.2 The Hamiltonian Constrains the Basis

A natural objection: who chose the projection? Why Hamming weight and not some other decomposition of the Hilbert space?

In our experiment, the choice was guided by two criteria from Strasberg's framework: the projector should be *coarse* (high rank, so that each subspace is large enough for the dynamics to spread correlations within it) and *compatible with the symmetry structure* of the Hamiltonian. The Hamming-weight threshold ≥ 4 satisfies both: it partitions $D = 512$

into subspaces of dimension 382 and 130, and total magnetization is a natural collective observable for an Ising chain.

Not every decomposition works. We verified that tracking a single qubit (two rank-256 projectors, $d_0 = d_1 = 256$) produces $\varepsilon \approx 1$ everywhere—no filtering, no stable branches. The reason is intuitive: a single qubit scrambles completely between steps, so successive projections carry no memory. The Hamming-weight projector averages over all 9 qubits, which smooths out the fast single-qubit fluctuations and preserves enough structure for the filtering to appear.

We did not systematically search over all possible coarse-grainings, so we cannot claim that Hamming weight is uniquely selected. But the contrast between "coarse projector shows filtering" and "fine projector does not" is consistent with Strasberg's framework, in which the Hamiltonian constrains which decompositions support approximately decoherent histories.

6.3 Anti-Born Branches: Not Just Unlikely, But Indistinct

The usual explanation for why we do not see anti-Born statistics is that those branches have low squared amplitude—they are "unlikely." The filtering mechanism suggests something additional. In our simulation, the anti-Born branches do not just have low weight—they also lose approximate decoherence, overlapping significantly with neighboring branches.

The all-zeros branch is a concrete example. Its weight is 2×10^{-8}, and its recoherence parameter is 0.77: it shares most of its overlap structure with neighboring branches. Whether this loss of decoherence constitutes a loss of "separate reality" depends on interpretive commitments we do not adjudicate here. But numerically, these branches fail the standard decoherent-histories criterion for classicality.

This observation—that anti-Born branches lose decoherence, not just weight—shifts the emphasis from "how likely is this branch?" to "does this branch maintain the approximate decoherence needed for stable records?"

6.4 Local Couplings

One might worry that the filtering pattern depends on the non-local, all-to-all coupling of random matrix Hamiltonians. Our Ising example provides one data point against this: the ZZ couplings are nearest-neighbor only, and the single-qubit fields are local. This is a baby lattice model with the same tensor-product structure $\mathcal{H} = \bigotimes_i \mathcal{H}_i$ that underlies lattice field theory. The fact that the qualitative filtering pattern appears with local gates is suggestive, though a single 9-qubit example obviously cannot establish generality.

6.5 Finite Dimensions as Physics, Not Limitation

Strasberg's mechanism requires finite D. This might seem restrictive, since textbook quantum field theory uses infinite-dimensional Fock spaces. But any lattice regularization of a quantum field theory lives in a finite-dimensional factored Hilbert space $\mathcal{H} = \bigotimes_x \mathcal{H}_x$, and physical arguments—the Bekenstein bound, holographic entropy bounds—independently suggest that the Hilbert space accessible to a bounded region is finite-dimensional. The mechanism does not require D to be small, only finite. A lattice gauge theory would provide the natural next test: finite D, local tensor-product structure, and a coarse-graining dictated by gauge invariance rather than chosen by hand.

6.6 Relation to Bohmian Mechanics

Bohm's pilot-wave theory [?] shares the single-thread intuition: one definite configuration is real at all times, and Born's rule emerges dynamically rather than by postulate. Two features were always seen as weaknesses: particle position is the preferred variable *by fiat*, and the guiding equation is an extra dynamical law beyond Schrödinger.

The recoherence picture, if it generalizes, removes both. The Hamiltonian selects the coarse-graining basis (Section 6): the preferred observable is derived, not chosen. And no guiding equation is needed—unitary evolution is the complete dynamics. In Bohm's framework, Born's rule requires assuming the initial particle distribution is $|\psi|^2$-distributed; in the recoherence framework, filtering is geometric—branches that violate Born statistics lose approximate decoherence regardless of the initial state.

Whether one thread is actual (as Bohm proposed) or all Born-typical threads are equally real (as Everett proposed) is not decided by the mechanism. But what Bohm had to postulate—a preferred variable and a statistical equilibrium—the finite geometry constrains.

6.7 The Remaining Question

Even if the filtering mechanism generalizes, it would not explain why *you* are in *this* particular branch. The mechanism, if it holds, would explain why the surviving branches have Born-typical statistics. It would not explain why you experience one branch rather than another. That question may be about indexical identity rather than physics, but we note it as an open issue rather than dismissing it.

But the picture, if it holds, is suggestive: the wave function evolves unitarily—as Everett proposed—yet only Born-typical branches remain approximately decoherent, and an observer would simply be one such branch. Collapse, in this reading, is not something that happens but the fact that you are one projection, experiencing one sequence of outcomes. If that framing survives scrutiny, then Copenhagen and Everett may both have been pointing at the same structure from different angles.

7 Conclusion

In a 9-qubit transverse-field Ising chain with exact unitary evolution, the recoherence parameter ε dips near the Born frequency with a $3.1\times$ gap over the combinatorial peak, matching the qualitative shape reported by Strasberg and collaborators in random matrix models. At the individual-branch level, the finite Hilbert space preferentially preserves Born-typical branches while anti-Born branches lose approximate decoherence. This is consistent with Strasberg's analytical results and extends the qualitative picture from random matrices to a local, physically motivated Hamiltonian.

A natural next step is to test the filtering mechanism in a lattice gauge theory, where the factored Hilbert space and local gauge constraints are physically motivated, and where the coarse-graining is dictated by the gauge symmetry—charge sectors, flux sectors—rather than chosen by hand. If the pattern survives when the projectors are fixed by gauge invariance rather than selected by the experimenter, the case for a physically grounded mechanism becomes considerably stronger.

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